SHUBHAM KHARCHE

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OBJECTIVE

AI Engineer with 2+ years of experience building scalable backend and Generative AI applications using Python, FastAPI, LangChain, and RAG pipelines. Skilled in developing LLM-powered systems, microservices, and data-driven APIs.

EXPERIENCE

Software Engineer — Oct 2025 - April 2026
InsightBridge - Heybobo.ai — *Bengaluru*

Module: AI-Powered Grooming
Tech Stack: FastAPI, OpenCV, MediaPipe, Deep Learning, CNN,Computer Vision, Microservices ,Image Processing

- Designed and developed a **scalable AI-driven grooming recommendation microservice** using Python, **enabling real-time analysis of user images** to deliver personalized skincare, hairstyle, and outfit suggestions
- Built **computer vision pipelines using OpenCV and MediaPipe** to perform facial landmark detection, skin condition classification, and body proportion analysis for accurate user attribute extraction.
- Developed and integrated **deep learning models (CNN / Vision Transformers)** for skin condition detection (acne, oiliness, dark circles) and generated dynamic health scoring metrics based on image analysis results.
- Engineered a **rule-based recommendation engine** that matches detected user attributes with structured product metadata, **improving recommendation accuracy** and personalization quality.
- Integrated AI-based recommendation logic with extensible architecture suitable for **LLM and Retrieval-Augmented Generation (RAG) pipelines** to enable future conversational grooming assistance

Module: AI-Powered Education
Tech Stack: FastAPI, NLP, LLM, RAG, FAISS, OCR, LangChain, OpenAI, MongoDB

- Designed and implemented a **scalable backend system** for an AI-powered education platform using Python and FastAPI, supporting personalized learning workflows, syllabus management, and **real-time student progress tracking.**
- Built a **Retrieval-Augmented Generation (RAG) pipeline using Sentence-BERT embeddings and FAISS vector indexing,** enabling semantic search and improving content retrieval latency and relevance for learning resources.
- Built **Generative AI-based question generation and homework assistance systems** using **T5 transformer** models and Large Language Models (LLMs) to automatically generate subject-specific questions and step-by-step explanations.
- Engineered student **performance analytics and classification models using Decision Trees and Random Forest algorithms,** enabling predictive insights and personalized difficulty adjustment based on learning behavior

Module: AI-Powered Intelligent Fitness
Tech Stack: FastAPI, NLP, LLM, RAG, LangChain, OpenAI, MongoDB

- Designed and developed scalable backend APIs using Python to **generate personalized workout plans** based on **user profile data including age, fitness level, injury history, and health conditions,** improving recommendation accuracy and user engagement.
- Implemented a **rule-based and data-driven workout plan generation engine** that dynamically adjusted exercise intensity and selection using injury validation logic and user progress metrics.
- Developed a **recommendation engine leveraging historical workout data** and user behavior patterns, simulating **RAG-style memory retrieval** to provide adaptive and non-repetitive workout suggestions.
- Designed and maintained **FastAPIs and modular service architecture**, supporting multi-sport fitness programs and enabling scalable feature expansion across cardio, strength training, and yoga modules.

Software Developer Intern — April 2024 - Sep 2024
InsightBridge — *Bengaluru*

Project: Dishoom- Ecommerce AI Aggregator

- Scraped and standardized product data from 200+ e-commerce websites across India, the USA, and the Middle East using **Selenium and Beautiful Soup**, validating API endpoints with Postman, and maintaining detailed documentation
- Designed and implemented a production-grade **RAG-based chatbot** for contextual product discovery and QA using document **chunking, Sentence Transformers Vector embeddings, ChromaDB**, and **OpenAI gpt-4o-mini** LLMs, with a user **feedback loop (like/dislike)** to track response quality and support iterative prompt and retrieval improvements.

- Built a **price history** analytics module with interactive **visualizations** and **timeframe filters**, integrating external ML services to generate **AI-powered Suggested Price** insights and statistical metrics such as average,median, and price ranges.

SKILLS

Programming Languages: Python, JavaScript, SQL

Frameworks & Libraries: FastAPI, Flask, Django, NumPy, Pandas, OpenCV, MediaPipe, React.js

AI / Machine Learning & Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Hugging Face Transformers, Sentence-BERT, Vector Embeddings, Prompt Engineering, Fine-tuning, Natural Language Processing (NLP), Computer Vision, Deep Learning, Convolutional Neural Networks (CNN), Vision Transformers, Recommendation Systems, Model Evaluation

Databases & Vector Stores: MongoDB, FAISS, ChromaDB, SQL Databases

APIs & Backend Development: REST APIs, Microservices Architecture, API Design, JSON, Asynchronous Programming, Backend Development

DevOps & Deployment: Docker, Git, GitHub, Linux, CI/CD, API Deployment, Model Deployment

Tools & Platforms: Postman, Jira, Visual Studio Code (VS Code), GitHub Actions

Concepts & Methodologies: System Design, Data Structures & Algorithms, Object-Oriented Programming (OOP), Agile Development, Software Development Lifecycle (SDLC)

PROJECTS

File Analyzer. Developed a comprehensive LangChain-powered file analysis application that enables users to upload PDF or TXT documents, automatically extracts and analyzes metadata such as page count, file size, character and paragraph counts, and optionally leverages an LLM to generate intelligent summaries and content interpretations for enhanced insights. (Link)

Text Predictor and Spell Checker. Developed a Markov chain–based text predictor and spell checker achieving 2ms sentence generation; evaluated and compared 1–5 gram language models.(Link)

EDUCATION

Master of Computer Applications, (MCA) 2021 - 2023
Savitribai Phule Pune University. CGPA: 7.88

Bachelor of Computer Applications, (BCA) 2018 - 2021
Kavayitri Bahinabai Chaudhari North Maharashtra University CGPA: 9.10

CERTIFICATIONS

- Develop AI Agents with LangChain LangGraph (udemy) - Link